



SMART IRRIGATION MONITORING SYSTEM



A Project Report

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BONAFIDE CERTIFICATE

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ABSTRACT

Water scarcity, climate variability, and inefficient irrigation practices have emerged as critical challenges in the agricultural sector, particularly in regions dependent on traditional farming methods. Conventional irrigation techniques, such as flood and manual irrigation, often result in excessive water usage, uneven distribution, soil degradation, and reduced crop productivity. Additionally, the lack of real-time monitoring and data-driven decision-making leads to poor water management, increased operational costs, and unnecessary human intervention. To address these issues, this project presents an Advanced, Intelligent, Sustainable, and Automated Smart Irrigation Monitoring System aimed at optimizing water utilization and enhancing agricultural efficiency through technology-driven solutions. The proposed system integrates a network of smart sensors, including soil moisture, temperature, and humidity sensors, along with a microcontroller-based control unit and wireless communication module. These components work together to continuously monitor environmental and soil conditions in real time. The collected data is transmitted to a centralized processing unit or cloud-based platform, where it is analysed using predefined algorithms and decision-making logic. Based on this analysis, the system automatically regulates the operation of irrigation pumps and control valves, ensuring that water is supplied precisely according to crop requirements. This closed-loop feedback control mechanism significantly reduces water wastage, prevents over-irrigation, and promotes efficient resource utilization. A key feature of the system is its automation capability, which minimizes the need for manual intervention and labour-intensive monitoring. Farmers can remotely access real-time data and control irrigation operations through a mobile application or web-based dashboard. This user-friendly interface provides alerts, graphical data visualization, and system status updates, enabling proactive decision-making and improved farm management. The system is also designed to be scalable and adaptable, making it suitable for small-scale farms, large agricultural fields, and greenhouse environments. From a sustainability perspective, the system contributes to responsible water management, conservation of natural resources, and reduction of energy consumption. By optimizing irrigation schedules and minimizing unnecessary water usage, it supports eco-friendly farming practices and aligns with global goals for sustainable agriculture and environmental protection. Furthermore, the intelligent nature of the system allows it to adapt to varying soil types, crop patterns, and climatic conditions, making it highly versatile and practical for real-world applications. In conclusion, the developed Advanced, Intelligent, Sustainable, and Automated Smart Irrigation Monitoring System offers a reliable, cost-effective, and technologically advanced solution for modern agriculture.

Keywords: Smart Irrigation; Internet of Things (IoT); Precision Agriculture; Automated Irrigation; Soil Moisture Monitoring; Water Conservation.

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LIST OF SYMOLS

SYMBOL	DESCRIPTION
V	Voltage (Volts)
A	Current (Amperes)
W	Power (Watts)
Ω	Resistance (Ohms)
°C	Temperature (Degrees Celsius)
%	Percentage (Humidity / Moisture Level)
MHz	Megahertz (Processor Clock Speed)
MB	Megabytes (Flash Memory)
L/min	Litres per Minute (Flow Rate)
DC	Direct Current
AC	Alternating Current

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
IoT	Internet of Things
ESP32	Espressis Systems Microcontroller 32-bit
DHT11	Digital Humidity and Temperature Sensor
LCD	Liquid Crystal Display
LED	Light Emitting Diode
GPIO	General Purpose Input Output
ADC	Analog to Digital Converter
I2C	Inter-Integrated Circuit
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver Transmitter
Wi-Fi	Wireless Fidelity
GSM	Global System for Mobile Communication
IDE	Integrated Development Environment
PCB	Printed Circuit Board
CAD	Computer Aided Design

LIST OF NOMENCLATURE

TERM	DESCRIPTION
Soil Moisture	Amount of water content present in the soil
Threshold Value	Predefined limit used for irrigation decision-making
Relay Module	Electrically controlled switch for pump operation
Microcontroller	Embedded processor unit (ESP32) controlling the system
Blynk Platform	IoT cloud platform for remote monitoring via mobile app
Closed-Loop Control	Automatic feedback-based control mechanism
Sensor Node	Device that senses and transmits environmental data
Prototype	Physical working model of the proposed system
Automation	Operation of system without manual intervention
Remote Monitoring	Monitoring system status from a distant location

CHAPTER 1

INTRODUCTION

Agriculture is the backbone of the economy and a primary source of livelihood for a large portion of the global population. The growth and productivity of crops are heavily dependent on the availability and proper management of water. However, rapid population growth, urbanization, industrial expansion, and climate change have significantly increased the demand for freshwater resources. As a result, water scarcity has become one of the most critical challenges faced by the agricultural sector today. In many regions, farmers rely on traditional irrigation methods such as flood irrigation, drip irrigation without automation, or manual watering, which often lead to inefficient water usage and resource wastage.

Conventional irrigation techniques lack precision and do not consider real-time soil and environmental conditions. Farmers usually irrigate their fields based on experience or fixed schedules rather than actual crop requirements. This practice results in over-irrigation in some areas and under-irrigation in others, leading to poor soil health, nutrient leaching, waterlogging, and reduced crop yield. Additionally, manual irrigation requires continuous human effort, time, and labor, making it inefficient and costly, especially for large agricultural lands.

To address these limitations, modern agriculture is gradually shifting toward smart and technology-driven solutions. The concept of smart irrigation integrates sensors, automation, and data analytics to create an efficient and sustainable water management system. In this context, this project proposes the development of an Advanced, Intelligent, Sustainable, and Automated Smart Irrigation Monitoring System that utilizes Internet of Things (IoT) and embedded systems technology to monitor and control irrigation processes in real time. The proposed system consists of soil moisture sensors, temperature and humidity sensors, a microcontroller-based control unit, and a wireless communication module. These components work together to continuously collect and analyse environmental and soil data from the field. Based on predefined threshold values and intelligent decision-making algorithms, the system automatically activates or deactivates irrigation pumps and valves as required. This closed-loop feedback control mechanism ensures precise water delivery, minimizes wastage, and maintains optimal soil conditions for plant growth.

Furthermore, the system allows remote monitoring and control through a mobile application or web-based dashboard, enabling farmers to access real-time field data from anywhere. Alerts and notifications can be sent to the user in case of abnormal conditions, such as extremely low soil moisture or pump failure. This feature enhances reliability, convenience, and overall farm management efficiency.

Overall, the proposed smart irrigation monitoring system aims to revolutionize traditional farming practices by promoting precision agriculture, conserving water resources, reducing manual labor, and improving crop productivity. It represents a significant step toward sustainable agriculture and smart farming, making irrigation more efficient, cost-effective, and environmentally responsible.

1.1 METHODOLOGY

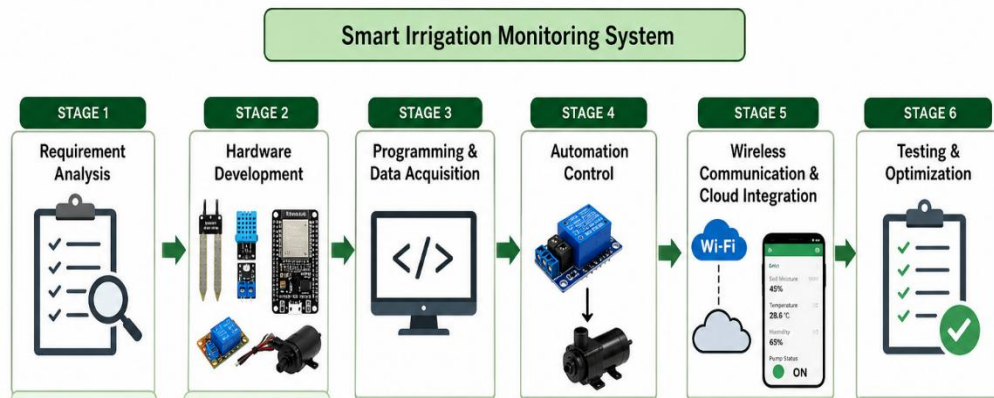


Figure 1.1 Methodology

Smart Irrigation Monitoring System involves a systematic approach that integrates hardware design, sensor data acquisition, data processing, automation control, and remote monitoring. The implementation is carried out in several stages to ensure accuracy, reliability, and efficiency of the system.

In the first stage, a detailed study of existing irrigation methods and smart irrigation technologies is conducted to identify their limitations and define the requirements of the proposed system. Based on this analysis, suitable sensors, microcontrollers, and communication modules are selected for system development.

In the second stage, the hardware components, including soil moisture sensor, temperature and humidity sensor, microcontroller (such as Arduino or ESP32), relay module, water pump, and power supply, are assembled and interconnected. The sensors are placed in the soil at appropriate depths to accurately measure soil moisture and environmental conditions.

In the third stage, the microcontroller is programmed using embedded C or Arduino IDE to read sensor data continuously. Threshold values for soil moisture are predefined based on crop and soil type. The microcontroller processes the data and makes intelligent decisions to control the irrigation pump automatically through a relay switch.

In the fourth stage, a wireless communication module (such as Wi-Fi or GSM) is integrated into the system to enable data transmission to a cloud platform or mobile application. This allows real-time monitoring of soil conditions and system status from a remote location.

In the final stage, the system is tested under different environmental and soil conditions to evaluate its performance, accuracy, and reliability. Necessary modifications are made based on test results to optimize system efficiency.

Overall, the adopted methodology ensures the successful design, implementation, and validation of an automated, intelligent, and sustainable smart irrigation monitoring system for modern agricultural applications.

CHAPTER 2

LITERATURE REVIEW

The development of smart irrigation systems has attracted significant attention from researchers due to increasing water scarcity, inefficient traditional irrigation practices, and the need for precision agriculture. Various studies have explored the use of sensors, automation, IoT, and data-driven decision-making to optimize water usage in agriculture. The following section presents a survey of relevant literature in the field of smart irrigation monitoring systems.

Ahmed et al. (2021)

Ahmed and his team developed a distributed IoT-based smart irrigation system for near real-time monitoring of soil moisture in the plant root zone. The system used soil moisture, air temperature, and relative humidity sensors, with cloud-based communication for monitoring and automated irrigation decisions. The study demonstrated the potential of IoT-based systems for improving irrigation management and reducing manual intervention.

Gore et al. (2021)

Gore et al. proposed a smart irrigation system integrating IoT and machine learning. The system monitored environmental parameters such as temperature and soil moisture and provided irrigation recommendations through an Android application. Machine learning techniques such as KNN were considered to improve irrigation decision-making based on environmental conditions.

Ndunagu et al. (2022)

Ndunagu and his team developed a wireless sensor network and IoT-based smart irrigation system using soil moisture and environmental parameters. The system used threshold-based irrigation control and cloud-based monitoring. Their evaluation demonstrated that IoT and machine learning can improve the reliability and efficiency of irrigation management.

Sangeetha et al. (2022)

Sangeetha et al. developed an IoT-based smart irrigation management system for environmental sustainability in India. The system integrated sensors and IoT technologies to monitor agricultural conditions and automate irrigation. The study focused on improving agricultural productivity while reducing unnecessary water consumption.

Togneri et al. (2022)

Togneri et al. investigated machine learning techniques for forecasting soil moisture in smart irrigation systems. Their study used IoT-generated soil moisture data collected from multiple farms and crop types. The results showed that machine learning models can support soil

moisture prediction and improve irrigation decision-making, although data quality remains an important challenge.

Anoop E. G. et al. (2023)

Anoop et al. proposed an IoT and machine learning-based automatic irrigation system for smart agriculture. The study analyzed monitoring parameters, hardware modules, communication technologies, and data storage methods to determine suitable irrigation operation. The work highlighted the role of IoT and ML in reducing water wastage and improving irrigation management.

Kranthi Kumar et al. (2024)

Kranthi Kumar et al. developed an intelligent irrigation system integrating IoT sensors with a Support Vector Machine approach. The research focused on using sensor information and machine learning for improved irrigation decision-making in agriculture. The study demonstrates the increasing integration of IoT sensing and machine learning techniques in intelligent irrigation systems.

2024 – Smart Irrigation for Tomato Cultivation

A comprehensive IoT and edge-computing-based irrigation system was developed specifically for tomato cultivation. The system used real-time field data to support irrigation scheduling and monitored parameters including water usage, soil nutrients, and plant height. The study reported improved efficiency and sustainability compared with conventional approaches.

Reddy et al. (2020)

The researchers proposed a low-cost IoT-based smart irrigation system for small-scale farmers. The system used affordable sensors and microcontrollers, making it accessible and practical for rural agricultural communities. The study emphasized that cost-effective smart irrigation solutions can significantly improve water management in developing countries.

Summary of Literature Review

The reviewed studies indicate that smart irrigation systems using sensors, automation, and IoT technologies have proven effective in reducing water wastage, improving crop productivity, and minimizing human intervention. However, challenges such as high cost, dependence on internet connectivity, system complexity, and maintenance requirements still exist. These limitations highlight the need for a simple, reliable, and cost-effective Advanced, Intelligent, Sustainable, and Automated Smart Irrigation Monitoring System, which is the focus of this project.

CHAPTER 3

COMPONENTS SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS SPECIFICATION TABLE

S.NO	COMPONENT	SPECIFICATION
1	ESP32 Wi-Fi Module	3.3V/5V, Dual-core, Inbuilt Wi-Fi
2	Soil Moisture Sensor	Analog output, 3.3V/5V
3	Temperature & Humidity Sensor (DHT11/DHT22)	Digital output, 3.3V/5V
4	Relay Module	5V, 10A
5	Water Pump (Motor)	12V DC, 3–5 L/min
6	LCD Display (16×2 / I2C)	5V
7	Green LED	2–3V
8	Red LED	2–3V
9	Power Supply	12V Adapter / Battery
10	Connecting Wires	Jumper wires
11	Blynk Platform	Web/Mobile App
12	Software	Arduino IDE

3.1 Components Specification

3.2 COMPONENTS DESCRIPTION

3.2.1 ESP32

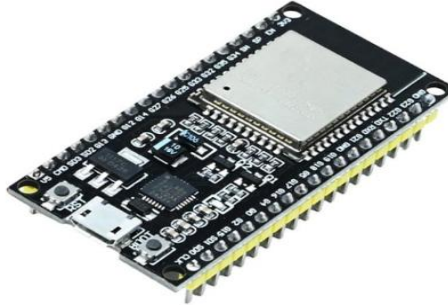


Figure 3.2.1 ESP32

The ESP32 is a low-cost, low-power, high-performance microcontroller with built-in Wi-Fi and Bluetooth capabilities, developed by Expressive Systems. It is widely used in Internet of Things (IoT) and automation applications due to its strong processing power, wireless communication, and multiple interfacing options. These features make it highly suitable for smart systems such as the Smart Irrigation Monitoring System.

The ESP32 is based on a dual-core Tensilica Xtensa LX6 processor operating at up to 240 MHz, which enables fast and efficient data processing. It supports low-power operation, making it suitable for battery-powered applications. The microcontroller also contains sufficient memory to handle sensor data, control logic, and wireless communication simultaneously.

One of the major advantages of the ESP32 is its integrated Wi-Fi (802.11 b/g/n) and Bluetooth, which allows real-time data transmission to cloud platforms and mobile applications such as Blynk or Thing Speak. In the proposed irrigation system, the ESP32 collects data from soil moisture and DHT11 sensors, processes it, and sends it to a mobile application for remote monitoring.

The ESP32 has multiple GPIO pins, ADC inputs, and communication interfaces like I2C, SPI, and UART, enabling easy connection with sensors, relay modules, LEDs, and display units. It can be programmed using the Arduino IDE, which makes it simple and user-friendly for students and researchers.

3.2.2 SOIL MOISTURE SENSOR

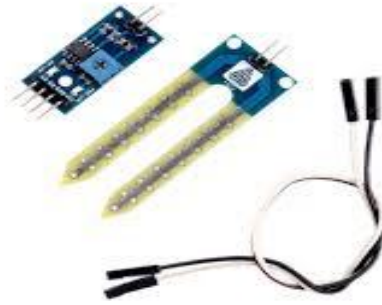


Figure 3.2.2 Soil Moisture Sensor

A soil moisture sensor is an essential component of the Smart Irrigation Monitoring System, used to measure the water content present in the soil. It helps in determining whether the soil is sufficiently wet or dry, allowing the system to make automated irrigation decisions. The sensor operates based on the principle of electrical conductivity between two probes inserted into the soil.

The sensor consists of two metal probes and a sensor module. When the soil contains more water, it becomes more conductive, resulting in a lower resistance and a higher sensor output value. Conversely, when the soil is dry, its resistance increases, leading to a lower output signal. This analog signal is sent to the ESP32 microcontroller for processing.

In the proposed system, the soil moisture sensor continuously monitors the moisture level and provides real-time data to the ESP32. Based on predefined threshold values, the microcontroller decides whether to turn the water pump ON or OFF through the relay module. If the moisture level is low, the pump is activated, and the red LED turns ON. If the moisture level is sufficient, the pump remains OFF, and the green LED turns ON.

The soil moisture sensor is easy to interface, cost-effective, and suitable for agricultural applications. It plays a crucial role in reducing water wastage, improving irrigation efficiency, and ensuring healthy crop growth through automated monitoring and control.

3.2.3 TEMPERATURE & HUMIDITY SENSOR (DHT11)



Figure 3.2.3 Temperature & Humidity Sensor

The Temperature and Humidity Sensor (DHT11/DHT22) is an important component in the Smart Irrigation Monitoring System used to measure the surrounding environmental conditions. This sensor provides digital readings of ambient temperature and relative humidity, which are useful for understanding climatic conditions affecting soil moisture and crop health.

The DHT11 sensor is a low-cost sensor with moderate accuracy, suitable for basic applications, while the DHT22 offers higher accuracy and a wider measurement range, making it more suitable for precise monitoring. Both sensors operate at 3.3V or 5V and communicate with the ESP32 microcontroller through a single digital data pin.

In the proposed system, the sensor continuously collects temperature and humidity data and sends it to the ESP32. The microcontroller processes this information and displays it on the LCD screen as well as uploads it to the Blynk web platform for real-time remote monitoring on a mobile phone.

3.2.4 RELAY MODULE



Figure 3.2.4 Relay Module

The relay module is a crucial component in the Smart Irrigation Monitoring System, used to control the ON/OFF operation of the water pump safely and effectively. Since the ESP32 operates at low voltage and current levels, it cannot directly drive a high-power device like a motor or pump. Therefore, the relay module acts as an electrically controlled switch between the low-voltage control circuit and the high-voltage pump circuit.

The relay module consists of an electromagnetic relay, a driver transistor, a diode for protection against back electromotive force (EMF), and an LED indicator for status visualization. When the ESP32 sends a HIGH or LOW signal to the relay input pin, the internal coil of the relay gets energized or de-energized accordingly. This action opens or closes the relay contacts, turning the water pump ON or OFF.

In the proposed system, when the soil moisture level falls below the predefined threshold, the ESP32 activates the relay module, which switches ON the water pump to supply water to the plants. Simultaneously, the red LED turns ON and the green LED turns OFF, indicating low moisture condition. When sufficient moisture is detected, the relay turns OFF the pump, the green LED turns ON, and the red LED turns OFF.

3.2.5 WATER PUMP (MOTOR)



Figure 3.2.5 Water Pump (Motor)

The water pump (motor) is a key actuator component in the Smart Irrigation Monitoring System, responsible for supplying water to the crops when irrigation is required. It is a 12V DC submersible or surface pump that draws water from a tank, reservoir, or water source and delivers it through pipes or drip lines to the agricultural field.

The pump operates based on control signals from the ESP32 through the relay module. Since the ESP32 cannot directly power the motor due to its low voltage and current limitations, the relay module acts as an intermediary switch to control the pump safely. When the relay is activated, the pump receives power from the external power supply and starts pumping water. When the relay is deactivated, the pump stops automatically.

In the proposed system, when the soil moisture sensor detects low moisture levels, the ESP32 sends a signal to the relay module, which turns ON the pump to irrigate the soil. At the same time, the red LED turns ON to indicate that watering is in progress. Once the desired moisture level is reached, the ESP32 turns OFF the relay, stopping the pump, and the green LED turns ON to indicate adequate moisture.

The water pump is energy-efficient, reliable, and suitable for small-scale and medium-scale irrigation systems. It plays a vital role in automating water distribution, reducing manual effort, and ensuring efficient water management in smart agriculture applications.

3.2.6 LCD DISPLAY



Figure 3.2.6 LCD Display

The 16×2 LCD display with I2C interface is used in the Smart Irrigation Monitoring System to visually display real-time sensor data and system status. The display consists of 16 columns and 2 rows, allowing it to show up to 32 characters at a time. It provides a simple and effective user interface for monitoring temperature, humidity, soil moisture level, and motor status without the need for a mobile phone or computer.

The I2C module reduces the number of required connections between the ESP32 and the LCD. Instead of using multiple data pins, it only requires two communication lines: SDA (data) and SCL (clock), along with power supply (VCC) and ground (GND). This simplifies wiring, saves GPIO pins on the ESP32, and makes the circuit more compact and organized.

In the proposed system, the LCD continuously displays the temperature, humidity, soil moisture percentage, and whether the motor is ON or OFF. For example, it may show messages like “Temp: 32°C”, “Humidity: 60%”, “Moisture: LOW”, and “Motor: ON”. This allows users to easily understand the system condition at a glance.

The LCD display enhances user interaction, improves system readability, and provides immediate feedback on irrigation performance. It is a reliable, low-power, and cost-effective component that plays an important role in the overall functionality of the smart irrigation system.

3.2.7 GREEN LED



Figure 3.2.7 Green LED

The Green LED is used as a visual indicator in the Smart Irrigation Monitoring System to represent a safe or adequate soil moisture condition. It provides an immediate and easily understandable status signal without requiring users to check the display or mobile application.

The Green LED is connected to a digital output pin of the ESP32 through a current-limiting resistor to prevent damage from excess current. When the ESP32 detects that the soil moisture level is above the predefined threshold (i.e., sufficient water content in the soil), it sends a HIGH signal to the Green LED, turning it ON.

In this condition, the relay remains OFF, the water pump does not operate, and the Green LED glows continuously to indicate that irrigation is not required. When the soil moisture level drops below the threshold, the Green LED automatically turns OFF, and the Red LED turns ON along with the motor.

The Green LED improves system usability by providing a clear, real-time visual indication of soil condition, making the system more user-friendly and effective for farmers and users.

3.2.8 RED LED



Figure 3.2.8 Red LED

The Red LED is used as a visual indicator in the Smart Irrigation Monitoring System to represent a low soil moisture or irrigation-required condition. It provides a clear and immediate warning signal to the user that the soil is dry and watering is in progress.

The Red LED is connected to a digital output pin of the ESP32 through a current-limiting resistor to protect it from excessive current. When the ESP32 detects that the soil moisture level is below the predefined threshold, it sends a HIGH signal to the Red LED, turning it ON.

In this condition, the relay module is activated, which switches ON the water pump to supply water to the soil. Simultaneously, the Green LED turns OFF, indicating that the soil is not in a safe moisture range. Once the soil reaches the required moisture level, the Red LED turns OFF, and the Green LED turns ON automatically.

The Red LED enhances the system's usability by providing a simple and intuitive visual alert, helping users quickly understand when irrigation is needed without checking the display or mobile application.

3.2.9 CONNECTING WIRES

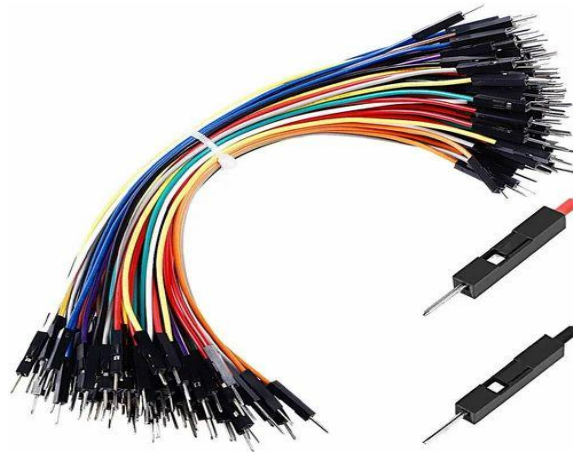


Figure 3.9 Connecting Wires

Connecting wires (jumper wires) are used in the Smart Irrigation Monitoring System to establish electrical connections between all components, including the ESP32, sensors, relay module, LCD display, LEDs, and power supply. These wires ensure proper signal transmission and power distribution throughout the circuit.

The system primarily uses male-to-male, male-to-female, and female-to-female jumper wires, depending on the type of connection required between the ESP32 and other components. For stable and secure connections, breadboard or soldered wiring can be used based on the prototype or final model design.

Proper wiring is essential to avoid loose connections, signal loss, or short circuits. The wires are arranged neatly to maintain circuit clarity, reduce interference, and improve system reliability. In the proposed setup, separate power and signal wires are used to minimize electrical noise and ensure accurate sensor readings.

Overall, connecting wires play a fundamental role in integrating all components into a functional smart irrigation system, enabling seamless communication and efficient operation.

3.2.10 BLYNK PLATFORM



Figure 3.10 Blynk Platform

Blynk is an Internet of Things (IoT) cloud platform used in the Smart Irrigation Monitoring System for real-time monitoring and control through a mobile application or web dashboard. It provides an easy-to-use interface that allows users to visualize sensor data and remotely manage the irrigation system from anywhere with internet connectivity.

In the proposed system, the ESP32 is connected to the Blynk platform via Wi-Fi. It continuously sends data such as soil moisture level, temperature, humidity, and motor status to the Blynk server. This data is displayed in the form of numerical values, gauges, or graphs on the user's mobile phone or web interface.

Blynk also enables remote control of the system. Users can manually turn the water pump ON or OFF using virtual buttons in the Blynk application if required. Additionally, alerts or notifications can be configured to inform users when soil moisture is low or when the pump is activated.

The Blynk platform enhances system flexibility, accessibility, and automation by enabling real-time data monitoring and remote irrigation control. It plays a crucial role in making the smart irrigation system more intelligent, user-friendly, and efficient for modern agricultural applications.

3.2.11 SOFTWARE

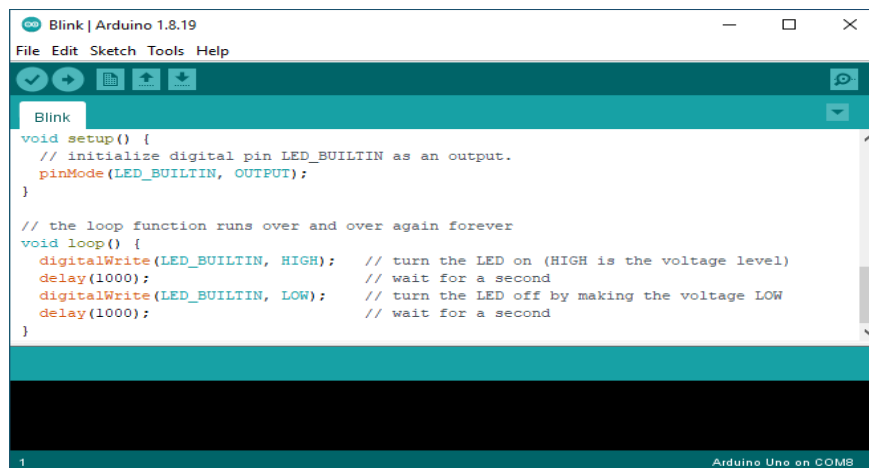


Figure 3.11 Software

The software plays a vital role in the Smart Irrigation Monitoring System by controlling data acquisition, decision-making, and communication between hardware components and the Blynk platform. The system is programmed using the Arduino IDE, which provides a simple and user-friendly environment for coding the ESP32.

The software continuously reads data from the soil moisture sensor and the DHT11 temperature and humidity sensor. Based on predefined threshold values, the program decides whether to turn the water pump ON or OFF through the relay module. It also controls the Green and Red LEDs to indicate soil moisture status and updates the LCD display with real-time sensor readings.

Additionally, the software enables Wi-Fi connectivity of the ESP32 and sends sensor data to the Blynk cloud platform for remote monitoring via mobile or web. The program also allows manual control of the motor through the Blynk application if needed.

Overall, the software integrates all system components, ensuring automation, real-time monitoring, and efficient irrigation management.

CHAPTER 4

WORKING PRINCIPLE

The Smart Irrigation Monitoring System is designed to optimize water usage in agriculture by automating the irrigation process based on real-time environmental and soil conditions. The system integrates sensors, a microcontroller, a water pump, and a cloud-based monitoring platform to create an efficient, reliable, and user-friendly irrigation mechanism. The primary objective of the system is to ensure that crops receive adequate water only when required, thereby preventing water wastage and improving agricultural productivity.

The working principle of the system is based on continuous monitoring of soil moisture, temperature, and humidity using appropriate sensors. These sensors are interfaced with a microcontroller (such as Arduino or ESP32), which acts as the central processing unit of the system. The soil moisture sensor is inserted into the soil at the root level of the plants to measure the water content present in the soil. It operates on the principle of electrical resistance or capacitance, where the moisture level in the soil affects the sensor's output voltage. When the soil is dry, the resistance is high, resulting in a higher sensor reading. Conversely, when the soil is wet, the resistance decreases, producing a lower sensor reading.

Along with soil moisture sensing, a temperature and humidity sensor (such as DHT11 or DHT22) is used to monitor the surrounding environmental conditions. This helps in understanding the climatic factors that influence soil dryness and plant water requirements. The sensor continuously collects data and sends it to the microcontroller in the form of digital or analog signals.

The microcontroller processes the incoming data using a pre-programmed algorithm. A predefined threshold value for soil moisture is stored in the system. If the measured soil moisture level falls below this threshold, the microcontroller interprets that the soil is dry and initiates the irrigation process. It sends a control signal to a relay module, which acts as an electronic switch to turn ON the water pump (motor). The pump then supplies water to the field through a network of pipes or drip irrigation emitters.

Once the soil moisture level reaches or exceeds the desired threshold, the microcontroller automatically turns OFF the pump by deactivating the relay. This closed-loop feedback mechanism ensures that water is supplied only when necessary, reducing manual intervention and conserving water resources. The entire process is automatic and does not require continuous human monitoring.

To enhance user interaction and monitoring, the system is integrated with an LCD display and the Blynk IoT platform. The LCD display provides real-time information such as soil moisture level, temperature, humidity, and pump status. This allows farmers or users to visually monitor the system's operation at the site. Additionally, the Blynk application enables remote monitoring and control through a smartphone or computer. The sensor data is transmitted to the cloud via Wi-Fi (using ESP32 or ESP8266), allowing users to check soil conditions from anywhere in the world.

The system also includes visual indicators such as a Green LED and Red LED. The Green LED indicates that the soil has sufficient moisture and the pump is OFF, while the Red LED indicates that the soil is dry and the pump is ON. This provides a simple and intuitive visual feedback mechanism.

In summary, the working principle of the Smart Irrigation Monitoring System is based on real-time data acquisition, intelligent decision-making, and automated control. By integrating sensors, microcontroller, relay, pump, and IoT connectivity, the system ensures efficient water management, reduces labor, prevents over-irrigation, and promotes sustainable agriculture. This technology is particularly beneficial in regions facing water scarcity and can significantly improve crop yield and resource utilization.

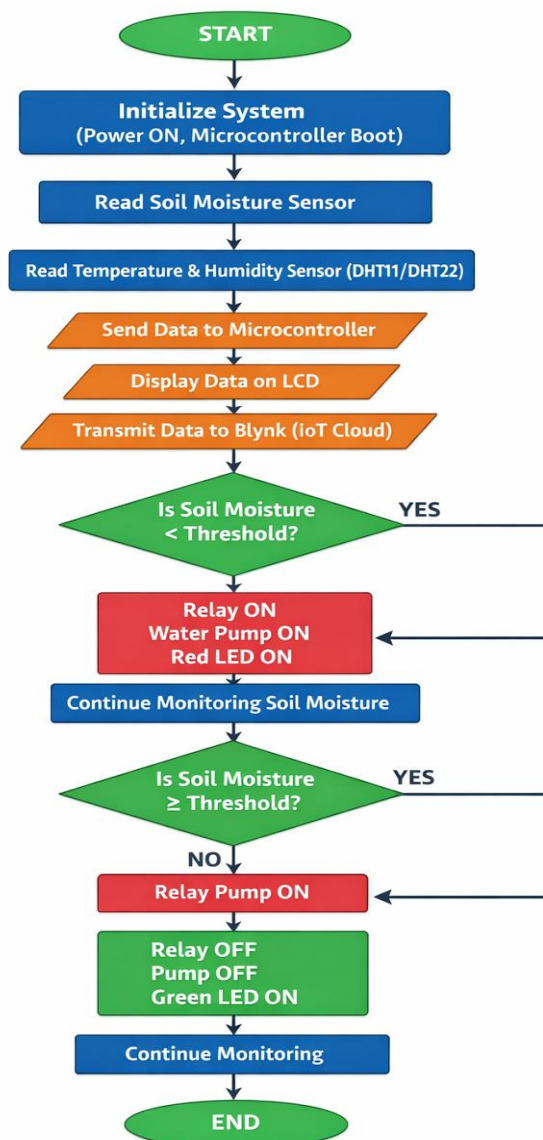


Figure 4.1 Flow Chart

CHAPTER 5

COST ESTIMATION

S.NO	Component Name	Specification	Quantity	Approx. Cost (₹)
1	Microcontroller	ESP32 (Wi-Fi & Bluetooth)	1	900
2	Soil Moisture Sensor	Resistive type (low-cost)	1	200
3	Temperature & Humidity Sensor	DHT11	1	120
4	Relay Module	5V Single Channel	1	100
5	Water Pump (DC Motor)	Mini 12V Pump	1	300
6	LCD Display	16×2 with I2C	1	300
7	Power Supply	12V Adapter	1	250
8	Green LED	Indicator	1	10
9	Red LED	Indicator	1	10
10	Connecting Wires	Jumper wires (set)	1	120
11	Pipes & Drip Tubes	Small irrigation setup	As required	150
12	Enclosure Box	Project casing	1	120
13	Blynk (IoT Platform)	Software	1	0 (Free)
Total				1480

5.1 Cost Estimation

CHAPTER 6

PROTOTYPE

A physical prototype was developed to demonstrate and validate the proposed Smart Irrigation Monitoring System under controlled conditions. The prototype represents a simplified working model of an automated irrigation setup incorporating soil moisture sensing, microcontroller-based decision making, and IoT-based remote monitoring features. The objective of this prototype is concept validation and system performance evaluation rather than large-scale field deployment. The prototype consists of an ESP32 microcontroller mounted on a breadboard, interfaced with a soil moisture sensor inserted into a soil sample container, and connected to a relay-controlled water pump. The sensor continuously measures soil moisture levels and transmits the data to the microcontroller for real-time processing.

When the soil moisture level falls below the predefined threshold, the microcontroller automatically activates the relay module, which switches ON the water pump to supply water. Simultaneously, the Red LED turns ON indicating irrigation in progress. Once adequate moisture is detected, the pump is turned OFF and the Green LED glows, confirming sufficient soil moisture. All sensor readings including temperature, humidity, soil moisture, and motor status are displayed on the 16×2 LCD screen and transmitted to the Blynk IoT platform for remote monitoring through a smartphone application.

6.1 3D CAD MODELING

Isometric view

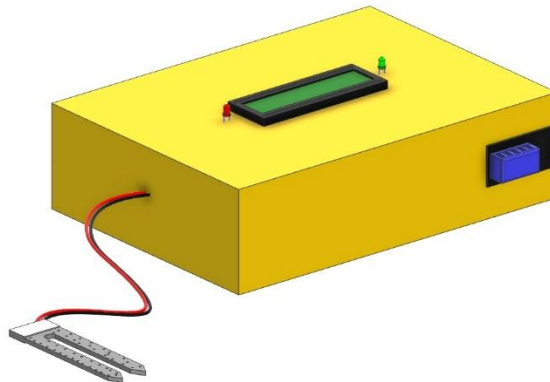


Figure 6.1.1 Isometric View

Top view

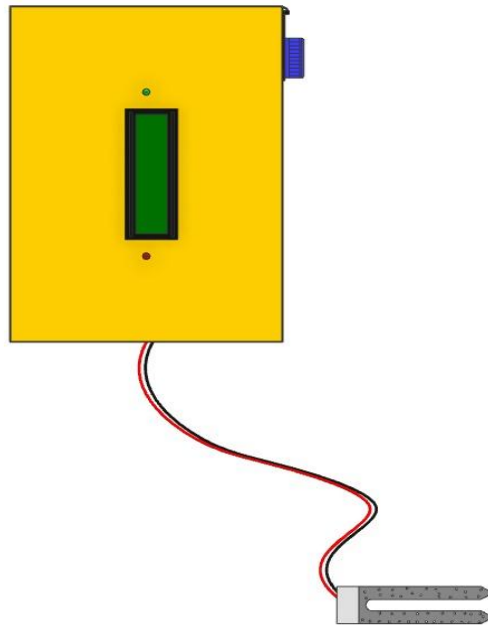


Figure 6.1.2 Top View

Front view

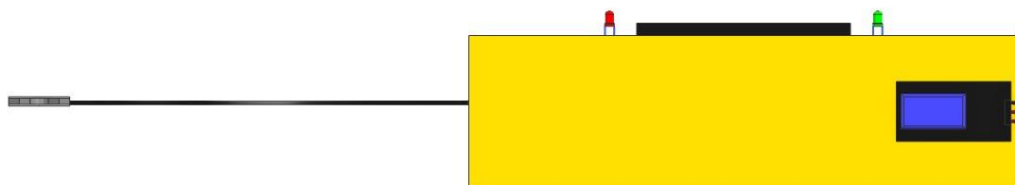


Figure 6.1.3 Front View

Right Side View

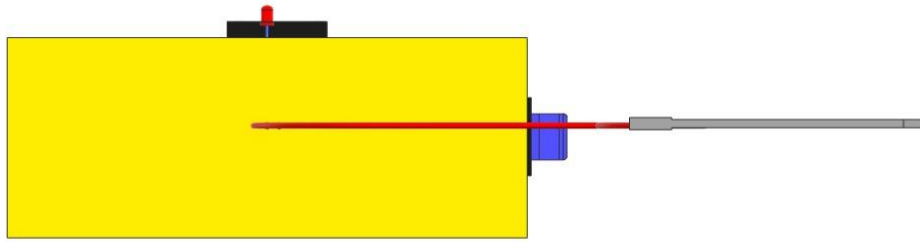


Figure 6.1.4 Right Side View

RESULTS DOCUMENTATION AND SYSTEM OUTPUTS

6.2 OUTPUT IMAGES

Output images represent the visual results obtained during the testing and operation of the Smart Irrigation Monitoring System. These images provide evidence of system performance and demonstrate the real-time working of hardware and software components. The images include sensor readings, pump operation status, LCD display outputs, and IoT monitoring results from the Blynk application. These outputs help to verify the accuracy and efficiency of the system.

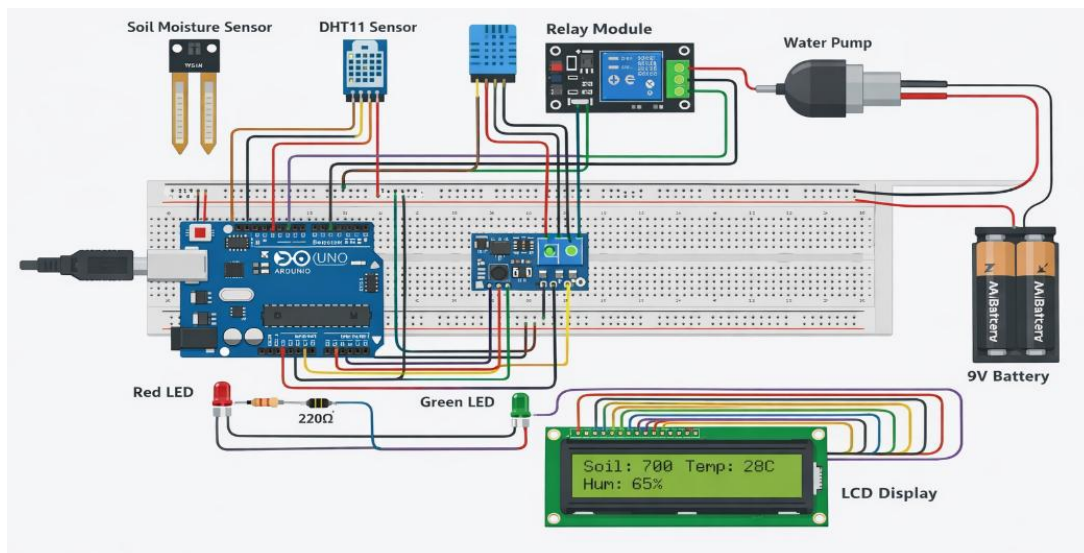


Figure 6.2 Simulation Model

- Hardware setup of the project
- LCD display showing moisture, temperature, humidity

- Pump ON condition (dry soil test)
- Pump OFF condition (wet soil test)
- Blynk mobile app output screen
- Soil moisture sensor testing images

6.2.1 Smart Irrigation Dashboard

The Smart Irrigation Dashboard is developed to provide real-time monitoring of the irrigation system through a web-based interface. The dashboard receives live sensor data from the ESP32 through the IoT platform and displays important parameters such as soil moisture, temperature, air humidity, and water pump status.

The Live Dashboard displays the current soil moisture percentage along with its target range. It also shows the temperature and humidity values obtained from the DHT11 sensor. The water pump status is displayed as ON or OFF based on the automatic irrigation control condition.

The dashboard also provides a Moisture Trend graph, which helps users observe the variation of soil moisture over time. The active watering profile displays the selected crop, soil type, pump ON threshold, pump OFF threshold, control mode, irrigation frequency, and total water used.

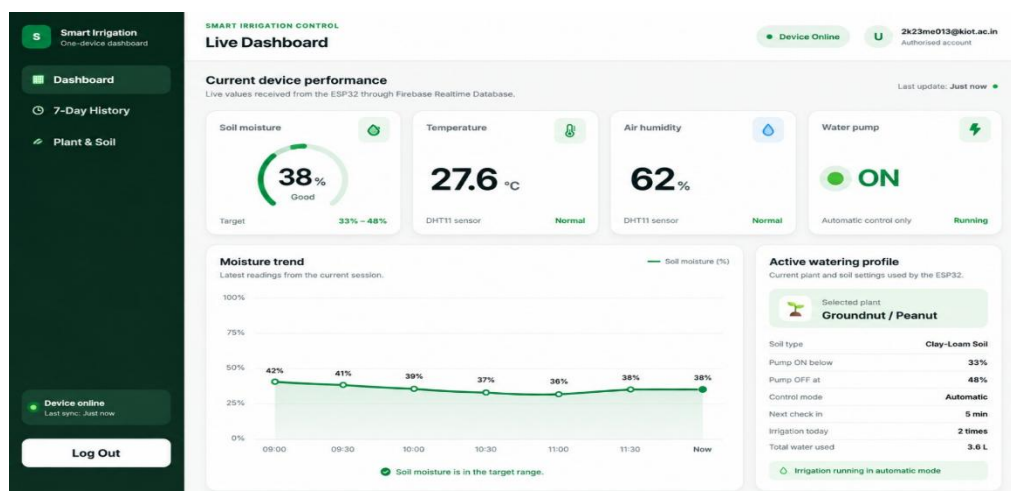
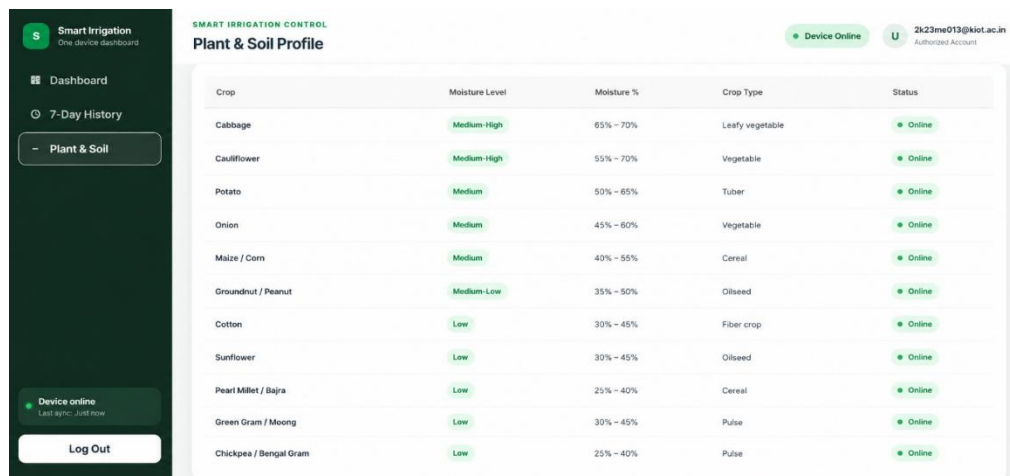


Figure 6.2.1 – Smart Irrigation Live Dashboard

6.2.2 Plant and Soil Profile

The Plant and Soil Profile page provides crop-specific soil moisture requirements. Different crops such as cabbage, cauliflower, potato, onion, maize, groundnut, cotton, sunflower, pearl millet, green gram, and chickpea are listed along with their required moisture ranges and crop types.

This feature helps the irrigation system select suitable moisture threshold values according to the crop requirement. It supports automatic irrigation management and prevents unnecessary watering.



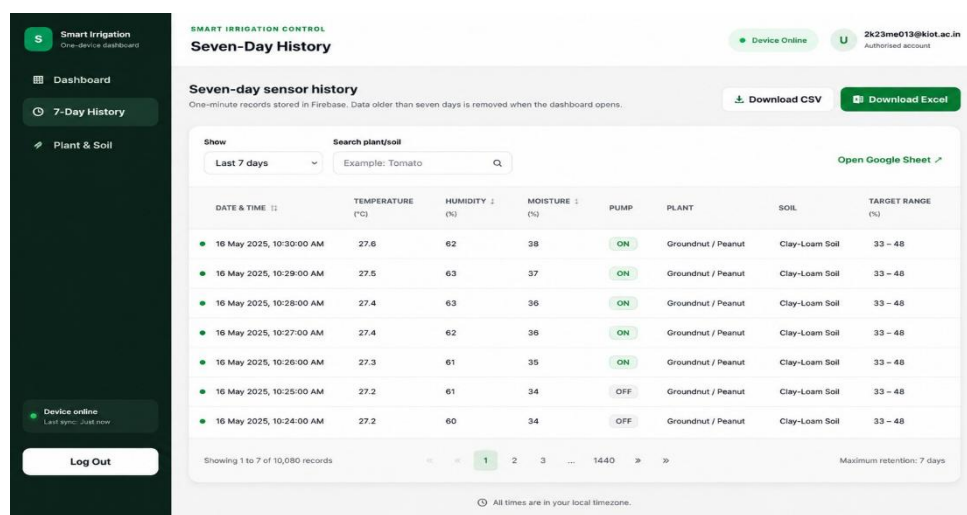
Crop	Moisture Level	Moisture %	Crop Type	Status
Cabbage	Medium-High	65% - 70%	Leafy vegetable	Online
Cauliflower	Medium-High	55% - 70%	Vegetable	Online
Potato	Medium	50% - 65%	Tuber	Online
Onion	Medium	45% - 60%	Vegetable	Online
Maize / Corn	Medium	40% - 55%	Cereal	Online
Groundnut / Peanut	Medium-Low	35% - 50%	Oilseed	Online
Cotton	Low	30% - 45%	Fiber crop	Online
Sunflower	Low	30% - 45%	Oilseed	Online
Pearl Millet / Bajra	Low	25% - 40%	Cereal	Online
Green Gram / Moong	Low	30% - 45%	Pulse	Online
Chickpea / Bengal Gram	Low	25% - 40%	Pulse	Online

Figure 6.2.2 – Plant and Soil Profile Dashboard

6.2.3 Seven-Day Sensor History

The Seven-Day History page stores and displays the sensor readings collected from the irrigation system. The recorded parameters include date and time, temperature, humidity, soil moisture, pump status, selected plant, soil type, and target moisture range.

The dashboard provides historical data for up to seven days and allows the user to search plant or soil information. The data can also be downloaded in CSV or Excel format for further analysis and documentation.



DATE & TIME	TEMPERATURE (°C)	HUMIDITY (%)	MOISTURE (%)	PUMP	PLANT	SOIL	TARGET RANGE (%)
16 May 2025, 10:30:00 AM	27.6	62	38	ON	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:29:00 AM	27.5	63	37	ON	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:28:00 AM	27.4	63	36	ON	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:27:00 AM	27.4	62	36	ON	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:26:00 AM	27.3	61	35	ON	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:25:00 AM	27.2	61	34	OFF	Groundnut / Peanut	Clay-Loam Soil	33 - 48
16 May 2025, 10:24:00 AM	27.2	60	34	OFF	Groundnut / Peanut	Clay-Loam Soil	33 - 48

Figure 6.2.3 – Seven-Day Sensor History Dashboard

6.2.4 Plant and Soil Configuration

The Plant and Soil Configuration defines the moisture requirement for different crops. It helps the system set suitable pump ON/OFF limits based on the selected crop and soil condition. Different crops have different moisture requirements, so the system uses crop-specific values for irrigation control. This helps maintain proper soil moisture for healthy plant growth. It also reduces over-irrigation, water wastage, and unnecessary pump operation.

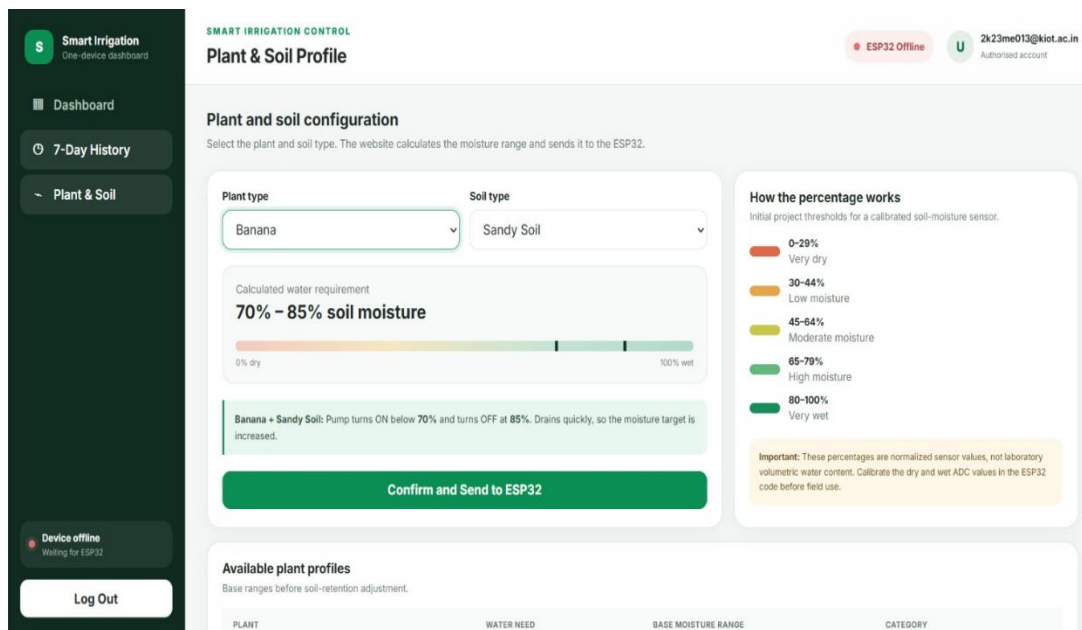


Figure 6.2.4 – Plant and Soil Configuration Dashboard

6.3 SYSTEM PERFORMANCE OBSERVATION

This section explains the observations made during system testing under different environmental and soil conditions. The system was tested by placing the soil moisture sensor in dry and wet soil samples. It was observed that when the soil moisture level dropped below the threshold value, the ESP32 microcontroller activated the relay and turned ON the water pump automatically. When sufficient moisture was detected, the pump was turned OFF, ensuring proper irrigation control.

6.4 RESULT TABLES

The result tables present the experimental readings obtained during testing of the Smart Irrigation Monitoring System. The readings include soil moisture, temperature, humidity, and pump status under different soil conditions.

Test No.	Plant Type	Soil Type	Soil Condition	Soil Moisture	Pump Status	Observation
1	Tomato	Loamy Soil	Dry	Low	ON	Irrigation required
2	Tomato	Loamy Soil	Wet	High	OFF	Moisture sufficient
3	Chilli	Sandy Soil	Dry	Low	ON	Irrigation required
4	Chilli	Sandy Soil	Moist	Medium	ON	Moisture maintained
5	Brinjal	Clay Soil	Wet	High	OFF	No irrigation required
6	Brinjal	Clay Soil	Dry	Low	ON	Irrigation required
7	Rose	Loamy Soil	Moist	Medium	ON	Moisture maintained
8	Spinach	Sandy Soil	Dry	Low	ON	Irrigation required
9	Coriander	Loamy Soil	Wet	High	OFF	Moisture sufficient
10	Marigold	Clay Soil	Moist	Medium	OFF	Irrigation not required

Table 6.1 – Plant, Soil and Irrigation Test Conditions

Result: The test results show that the pump automatically turns ON when soil moisture is low and turns OFF when sufficient moisture is detected. This confirms the automatic irrigation control of the proposed system.

CHAPTER 7

SCOPE OF WORK

7.1 SCOPE OF FUTURE WORK

- Integration of advanced sensors such as pH sensors, nutrient sensors, and rain sensors for better soil and environmental monitoring.
- Implementation of weather forecasting integration to improve irrigation scheduling.
- Development of wireless sensor networks for monitoring large agricultural fields.
- Expansion of the system to support multiple crop monitoring and automatic irrigation customization.
- Improvement in sensor durability by replacing resistive sensors with capacitive moisture sensors.
- Integration of automatic fertilizer supply system along with irrigation control.

7.2 ADVANTAGES

- Reduces water wastage by automatic irrigation control.
- Minimizes manual labour and human effort.
- Improves crop growth by maintaining proper soil moisture.
- Provides real-time monitoring using IoT technology.
- Enables remote monitoring and control through mobile applications.
- Promotes water conservation and sustainable agriculture.

7.3 APPLICATIONS

- Agricultural fields for automatic irrigation.
- Greenhouses and polyhouse farming.
- Home gardens and smart gardening systems.
- Nurseries and plant cultivation centres.
- Landscaping and lawn irrigation.

CHAPTER 8

APPENDICES

APPENDIX: 1 - FIELD VISIT

As part of the Smart Irrigation Monitoring System project, a field visit was conducted to study the existing irrigation practices and understand real-time agricultural challenges. The visit was carried out in a nearby agricultural farm where traditional irrigation methods such as manual watering and motor-based pumping were being used. During the visit, observations were made regarding water usage, soil conditions, crop type, and farmer involvement in irrigation management.

It was observed that irrigation was mainly performed manually or based on fixed time intervals without considering the actual soil moisture condition. This often resulted in overwatering or underwatering, leading to water wastage and uneven crop growth. The farmers also faced difficulties in continuously monitoring soil conditions, especially during hot climatic conditions. Additionally, electricity consumption for water pumps was found to be higher due to the absence of automatic control mechanisms.

Based on these observations, the need for an automated irrigation monitoring system was clearly identified. The Smart Irrigation Monitoring System was designed to address these challenges by incorporating soil moisture sensors, environmental monitoring, and IoT-based remote control using ESP32. The system ensures that water is supplied only when required, thereby conserving water, reducing manual labour, and improving crop productivity.

The field visit helped in understanding practical issues faced in agriculture and provided valuable insights for designing an efficient and user-friendly irrigation solution. It also confirmed the feasibility and real-world applicability of the proposed system.



Figure 8.1 Field Visit



Figure 8.2 Prototype Testing

APPENDIX – 2

COMPONENTS SPECIFICATION

PROTOTYPE SPECIFICATION

1. ESP32 Microcontroller

- Operating Voltage: 3.3V
- Input Voltage (Recommended): 5V–12V
- Wi-Fi: 802.11 b/g/n
- Bluetooth: v4.2 (BLE + Classic)
- CPU: Dual-core 32-bit processor
- Clock Speed: Up to 240 MHz
- Digital I/O Pins: 30+
- Analog Input Pins: 12-bit ADC
- Flash Memory: 4MB
- Communication Protocols: UART, SPI, I2C

2. Soil Moisture Sensor

- Operating Voltage: 3.3V – 5V
- Output Type: Analog & Digital
- Measuring Range: 0–100% moisture level
- Material: Corrosion-prone metal probes

3. DHT11 Temperature & Humidity Sensor

- Operating Voltage: 3V – 5V
- Temperature Range: 0°C to 50°C
- Temperature Accuracy: $\pm 2^{\circ}\text{C}$
- Humidity Range: 20% – 90% RH
- Humidity Accuracy: $\pm 5\%$
- Output: Digital Signal

4. Relay Module

- Operating Voltage: 5V
- Trigger Voltage: 3.3V/5V compatible
- Type: Electromechanical relay
- Load Capacity:
 1. 10A at 250V AC
 2. 10A at 30V DC

5. 12V DC Water Pump

- Operating Voltage: 12V DC
- Current Consumption: 0.5A – 1A
- Power Rating: 6W – 12W
- Flow Rate: 2–5 liters/min
- Type: Submersible mini pump

6. LCD Display

- Operating Voltage: 5V
- Display Type: Alphanumeric
- Characters: 16 characters × 2 lines
- Communication: I2C protocol
- I2C Address: 0x27

7. LEDs (Red & Green)

- Operating Voltage: 2V – 3V
- Current: 20Ma
- Red LED → Pump ON indicator
- Green LED → Pump OFF indicator

8. Connecting Wires

- Type: Male-to-Male / Male-to-Female
- Length: 10–20 cm
- Used for breadboard connections

APPENDIX – 3

PSEUDO CODE FOR KEY FUNCTIONS

Pseudo Code 1: System Initialization Function

START System Initialization

Initialize serial communication

Configure RELAY_PIN as OUTPUT

Configure RED_LED as OUTPUT

Configure GREEN_LED as OUTPUT

Set RELAY_PIN to OFF

Set RED_LED to OFF

Set GREEN_LED to ON

Initialize I2C communication

Initialize LCD display

Turn LCD backlight ON

Initialize DHT temperature and humidity sensor

Display "WELCOME" on LCD

Display "TEAM AEROLEAF"

Wait for 2 seconds

Clear LCD

Display "SENSOR"

Display "COLLECTING DATA"

Wait for 2 seconds

END System Initialization

Pseudo Code 2: Soil Moisture Reading Function

START Read_Soil_Moisture

Read analog value from SOIL_PIN

Store value in soil Value variable

RETURN soil Value

END Read_Soil_Moisture

Pseudo Code 3: Temperature and Humidity Reading Function

START Read_Temperature_Humidity

Read humidity from DHT sensor

Read temperature from DHT sensor

IF humidity OR temperature is invalid

Set humidity = 0

Set temperature = 0

END IF

RETURN temperature, humidity

END Read_Temperature_Humidity

Pseudo Code 4: Irrigation Control Function

START Irrigation Control

IF soilValue > soilDry AND temperature > tempLimit AND humidity < humLimit

 Turn RELAY ON

 Turn RED_LED ON

 Turn GREEN_LED OFF

 motorState = ON

ELSE

 Turn RELAY OFF

 Turn RED_LED OFF

 Turn GREEN_LED ON

 motorState = OFF

END IF

RETURN motorState

END Irrigation_Control

Pseudo Code 5: LCD Display Function

START Display_Data

Clear LCD

Display temperature on first row

Display humidity on first row

Display soil moisture value on second row

IF motorState = ON

 Display "ON"

ELSE

 Display "OFF"

END IF

END Display_Data

Pseudo Code 6: Main Monitoring Loop

START Main_Loop

WHILE system is running

 soilValue = Read_Soil_Moisture()

 temperature, humidity = Read_Temperature_Humidity()

```
motorState = Irrigation_Control()
```

```
Display_Data()
```

```
Wait for 2.5 seconds
```

```
END WHILE
```

```
END Main_Loop
```

CHAPTER 9

CONCLUSION

The Smart Irrigation Monitoring System was successfully designed and implemented using an ESP32 microcontroller, soil moisture sensor, DHT11 temperature and humidity sensor, relay module, and IoT-based monitoring through the Blynk platform. The primary objective of this project was to develop an automated irrigation system that efficiently manages water usage based on real-time soil and environmental conditions. The developed prototype effectively monitors soil moisture levels and automatically controls the water pump when the moisture level falls below a predefined threshold. This ensures that crops receive adequate water without over-irrigation or wastage. The integration of Wi-Fi connectivity enables remote monitoring and control through a smartphone application, making the system more convenient and user-friendly. During testing, the system demonstrated accurate sensor readings, quick response time, and reliable pump operation. The automatic cut-off mechanism successfully prevented excessive water supply, thereby conserving water and reducing manual effort. The system is suitable for small-scale agricultural fields, gardens, and greenhouse applications. Overall, the Smart Irrigation Monitoring System provides a cost-effective, efficient, and sustainable solution for modern agriculture. It reduces water consumption, minimizes labor requirements, and promotes smart farming practices. With further enhancements such as additional sensors and large-scale implementation, the system can significantly contribute to improved agricultural productivity and resource management.

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